

Efficacy and Safety of FDC of Dapagliflozin, Glimepiride and Metformin ER Tablets in Indian T2D Patients: A Subgroup Analysis of Patients with Generalized Obesity (BMI ≥ 25 kg/m²)

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Sahay Rakesh¹, Kurmi Prakash², Ambaliya Jayesh³, Lomte Nilesh⁴, **Dharmadhikari Shruti^{5*}**, Rajurkar Mandodari⁵, Saha Shruti⁵, Patil Dipak⁵, Ghadge Pravin⁵, Lakhwani Lalit⁵, Mehta Suyog⁵, Joglekar Sadhna⁶, Dapagliflozin, Glimepiride and Metformin extended-release study group⁷

*Presenting author; 1. Osmania Medical College, Hyderabad, India; 2. Shivam Medical Hospital, Ahmedabad, India; 3. Pagarav Hospital, Gandhinagar, India; 4. Hormone Care, Aurangabad, India; 5. Sun Pharma Laboratories Ltd, Mumbai, India 6. Sun Pharmaceutical Industries Ltd, Mumbai, India 7. Multiple locations, India

Background

- The prevalence of generalized obesity (BMI ≥ 25 kg/m²) is 28.6% among Indian diabetics¹
- Metformin, as a first-line agent, has demonstrated improved CV outcomes²
- Glimepiride, a modern sulfonylurea, is a CV-neutral agent with lesser CV side effects compared to conventional sulfonylureas³
- Extensive clinical data favour Dapagliflozin in reducing HbA1c while mitigating CV risk and renal complications⁴
- Combining Dapagliflozin with a novel mechanism of action & beta cell sparing effects with Metformin and Glimepiride can help achieve aggressive glycemic goals by enhancing HbA1c reduction compared to FDC of Glimepiride and Metformin in T2D patients with generalized obesity.

Aim

To evaluate the efficacy and safety of FDC of Dapagliflozin, Glimepiride and Metformin HCl ER tablets in comparison to FDC of Metformin HCl PR and Glimepiride tablets in Indian patients with T2D. A subgroup analysis of patients with BMI ≥ 25 kg/m² at 16 weeks is presented.

Material and methods

Study design

A multicenter, randomized, active-controlled, open-label, parallel-group, two-arm, phase III study

Primary endpoint

Mean change in HbA1c from baseline to Week 16

Secondary endpoints

- Participants achieving HbA1c < 7.0% at the end of Weeks 12 and 16
- Treatment-emergent adverse events (TEAEs)

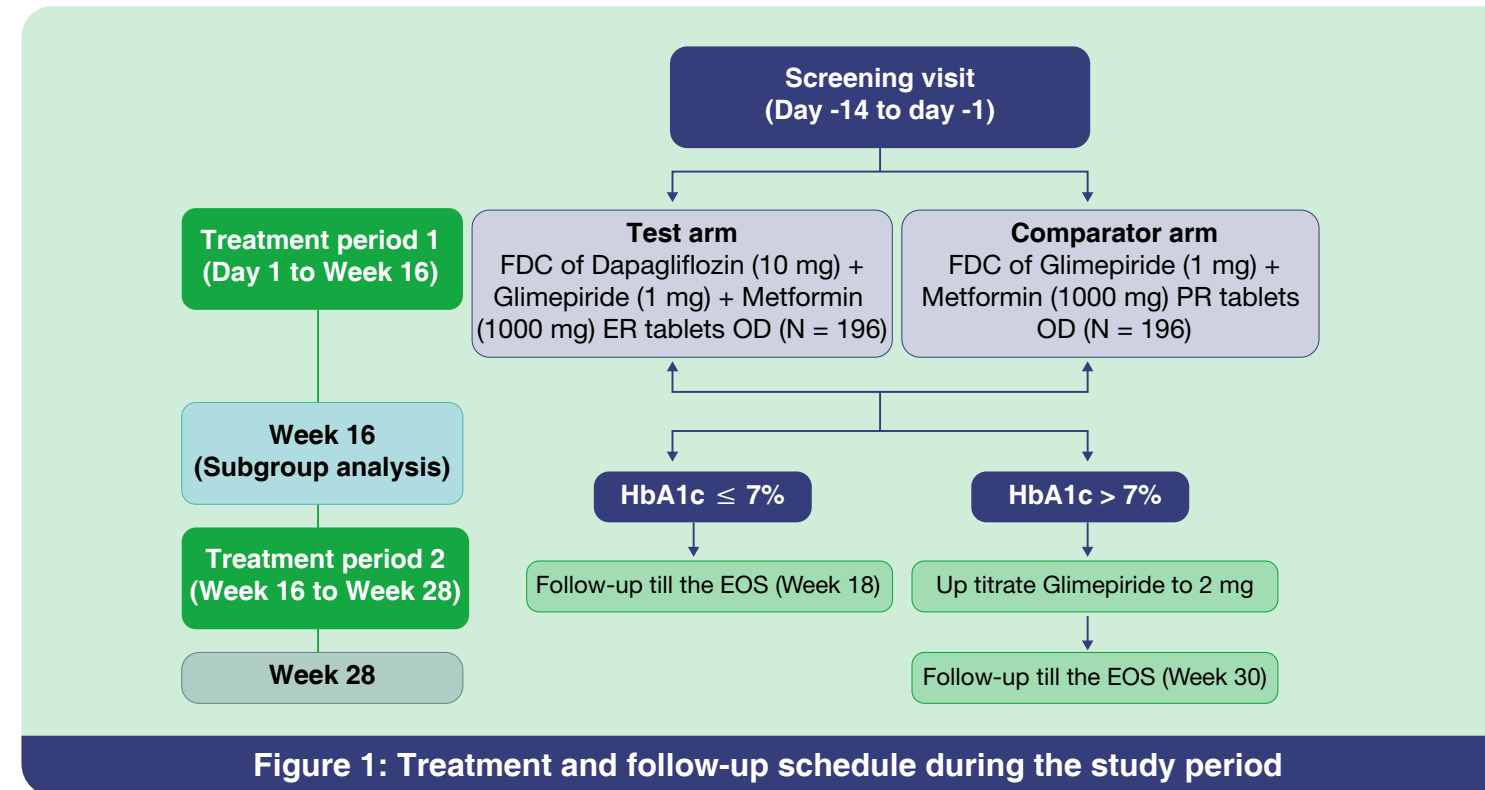
Inclusion criteria

T2D patients of either gender, aged 18–65 years

- HbA1c $\geq 8\%$ – $\leq 11\%$
- BMI ≤ 45 kg/m²
- Lifestyle modifications and stable daily dose of Glimepiride 1 mg and Metformin SR/PR/ER 1000 mg for at least 8 weeks prior to screening

Exclusion criteria

- FBG ≥ 270 mg/dL at screening
- Any weight loss medications within 3 months prior to randomization
- Significant renal impairment (eGFR < 45 mL/min/1.73 m²) or hepatic impairment (AST and AST > 3 times upper limit of normal)
- Uncontrolled hypertension $\geq 160/100$ mmHg



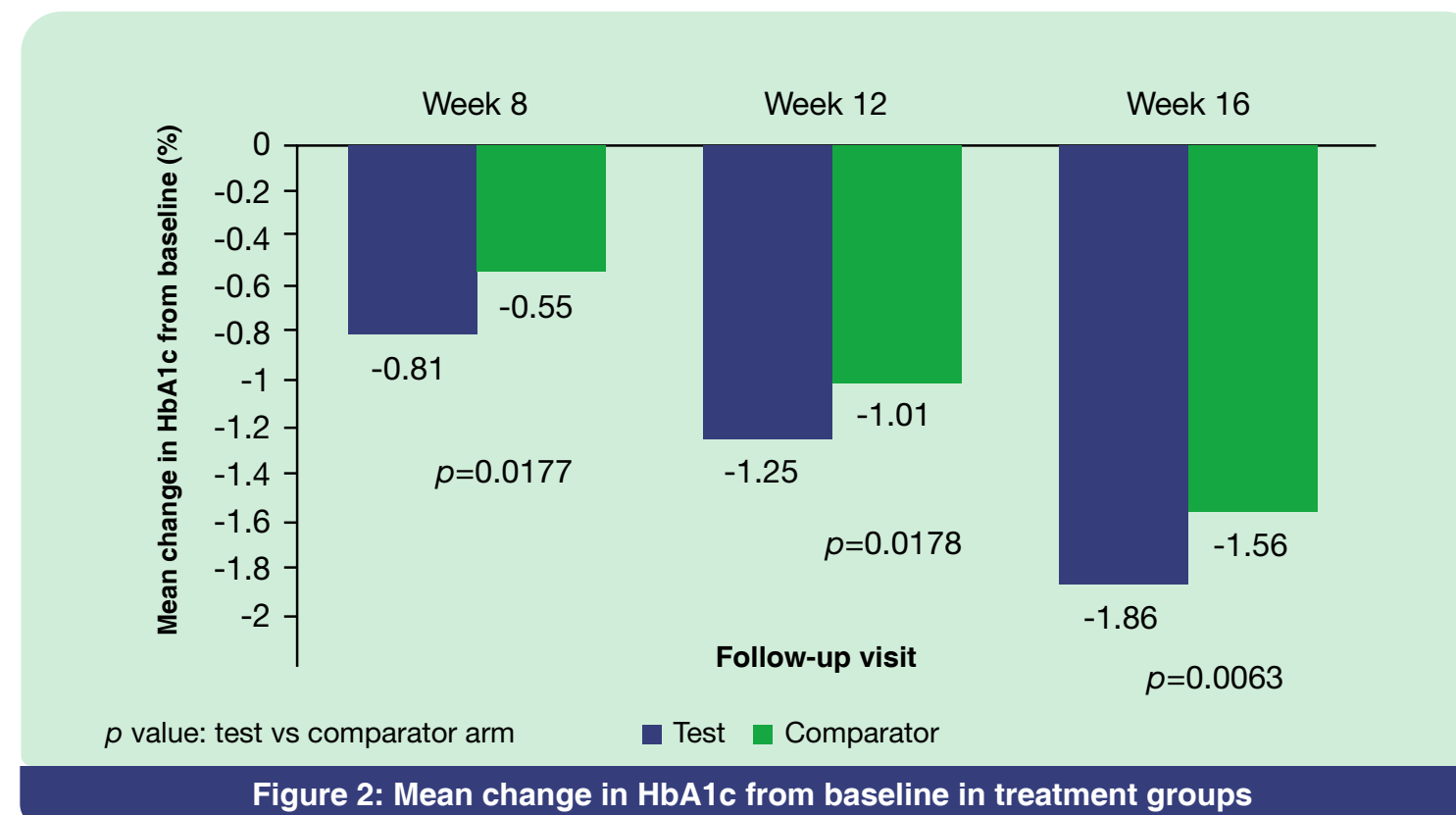
Results

Baseline characteristics

- Patients with generalized obesity *BMI ≥ 25 kg/m² = 217 (test = 104; comparator = 113)
- HbA1c
 - Test = 8.99% $\pm$ 0.83%
 - Comparator = 8.95% $\pm$ 0.72%

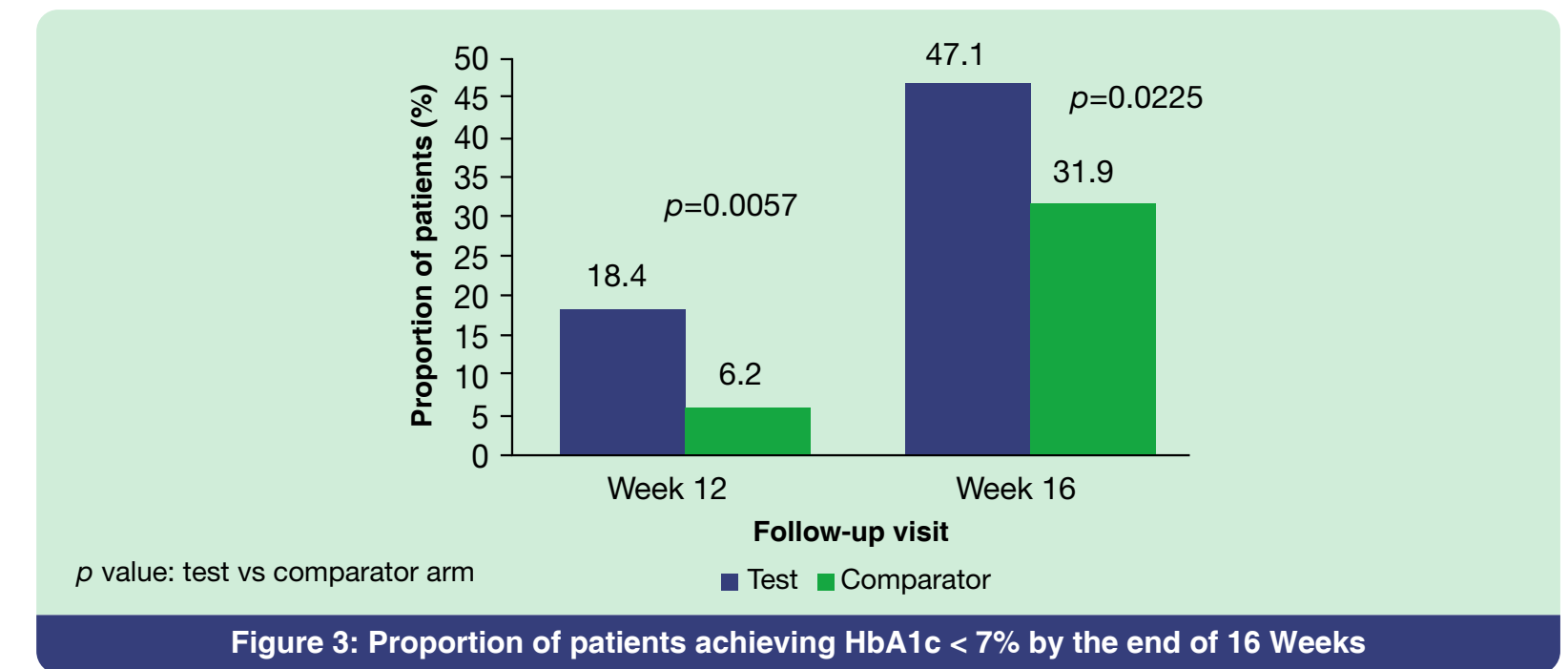
Mean change in HbA1c at Week 16

The mean change in HbA1c was statistically significant in the test arm vs. comparator arm ($p=0.0063$, Figure 2)



Proportion of patients achieving HbA1c < 7% at Week 16

Significantly more patients from the test arm achieved HbA1c level < 7% in comparison with the comparator arm at Weeks 12 and 16 (Figure 3)



TEAEs

- Across different age groups, TEAEs were reported in 8 subjects from the test arm and 7 from the comparator arm. These events were of mild intensity and did not necessitate treatment interruption, discontinuation, or dose reduction or result in fatalities
- In both the test and comparator arms, one TEAE was likely related, while the remaining events were unlikely to be related
- The study medications were well-tolerated, with no SAEs reported

Conclusion

- In the subgroup of patients with a BMI ≥ 25 kg/m², the triple FDC of Dapagliflozin, Glimepiride and ER Metformin Hydrochloride tablets was superior in comparison to FDC of Metformin Hydrochloride PR and Glimepiride tablets in terms of HbA1c reduction at the end of Week 16
- The proportion of patients achieving HbA1c < 7.0% was higher and statistically significant in the test arm compared to the comparator arm at the end of Week 16, demonstrating better glycemic control
- Results of safety analysis showed that the incidence of TEAEs was comparable in both the arms
- Overall, the study products were safe and well-tolerated

Abbreviations

ALT, Alanine transaminase; AST, Aspartate aminotransferase; BMI, Body mass index; CV, Cardiovascular; CVD, Cardiovascular disease; D, day; EOS, End of study; eGFR, Estimated glomerular filtration rate; ER, Extended release; FBG, Fasting blood glucose; FDC, Fixed dose combination; HbA1c, Glycated haemoglobin; HCl, Hydrochloride; OD, Once daily; PPBG, Postprandial blood glucose; PR, Prolonged release; SAE, Serious adverse event; SR, Sustained release; T2D, Type 2 diabetes; TEAE, Treatment emergent adverse event

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